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The End of the Universe

Background:

Kurzgesagt – The Last Thing To Every Happen In The Universe

- 90% of all stars that will ever exist have already been born
- Eventually all stars will become white dwarves, neutron stars, or black holes as the universe "runs out of gas"
- The mass of galaxies at this point will be:
  - 88% white dwarves
  - 10% gas giants and brown dwarves
  - 2% neutron stars and black holes

White Dwarfs

- Hot, dim, dense spheres of carbon
- Loses heat *very* slowly
- After at least 10 trillion years, the white dwarves cool down to just above absolute 0 and become *black dwarves*
- After 10^100 years, all galaxies have evaporated, leaving the remaining bodies to drift on their own

Black Holes

- All black holes emit *hawking radiation* which causes it to slowly lose mass

- After 10^100 years, the last black hole completely evaporates, leaving black dwarves the final astronomical objects in the universe

Black Dwarfs

- Carbon nuclei are compressed under immense pressure into a lattice structure
- The pressure results in electrons being unable to bond properly to these nuclei, so they become a plasma that fills in the gaps between nuclei.
- The repulsive force between these free electrons is what counters the gravitational force compressing the star radially inward

Quantum Tunneling

- Quantum Tunneling is a phenomenon in quantum mechanics where a particle passes through an energy barrier despite not having enough energy to do so
- Quantum tunneling causes very rare and isolated cases of fusion to occur within the black dwarf
- The nuclei inside the black dwarf continue to fuse until all the silicon fuses into Nickel-56, a radioactive isotope of nickel.
- Nickel-56 decays into iron and emits 2 positrons
- Energy cannot be extracted from iron
- The positrons collide with 2 electrons and totally annihilate them

Heat Death

- Eventually, there will not be enough electrons left to hold counter the force of gravity, resulting in a supernova

- When this happens to the last black dwarf and the final particle has cooled to absolute 0, the heat death of the universe has been reached and nothing can ever happen again

Kurzgesagt – How to Destroy the Universe

- As the universe expands, new space is being created out of nothing
- This empty space appears empty, but carries an inherent energy called Dark Energy
- The force of gravity wants to pull objects together
- Dark energy counters this force and keeps the universe expanding and accelerating
- The question is whether this will go on forever or will dark energy get weaker over time
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Heat Death

- This hypothesis assumes dark energy's strength remains constant
- As the universe expands, more and more dark energy is created
  - This in turn accelerates the universes expansion outwards
  - The universe will double in "diameter" every 12 billion years
- Eventually, the expansion becomes more noticeable at a scale observable by humans:
  - Galactic super clusters will expand away from each other
  - Then galaxies will be pulled apart
  - Then solar systems
- By this point, the universe contains only black holes and black dwarfs moving farther away from each other
- After 10^{100} years, the last black holes evaporate and the universe has experienced heat death

The Big Rip

- This hypothesis assumes that the strength of dark energy increases over time
- This would exponentially speed up the expansion of the universe
- This would also cause empty space to be created in smaller and smaller regions
    - First at a galactic level much like the heat death theory
    - ~1 billion years after this, space will begin appearing between stars, causing galaxies to dissolve
    - A few million years after that, space begins appearing inside star systems, pushing planets away from their stars.
    - This will continue until atoms are torn apart by space being created between electrons and nuclei
    - A few nanoseconds later, space time itself is broken completely and we can no longer predict what would happen
    - This is called the Big Rip

The Big Crunch

- This hypothesis assumes that dark energy will get weaker over time
- The result of this is that the force of dark energy being overwhelmed by the force of gravity
- This would cause all the mass in the universe to begin moving towards each other
- If space itself shrinks, then all the leftover radiation from the big bang and every quasar and supernova to begin to blue shift
- The opposite of the big rip occurs
    - Planets crash into their stars
    - Galaxies collapse in on themselves

- Eventually all the matter in the universe is collapsed into a single ball of plasma much like just after the big bang

- All the matter in the universe will be crushed into a singularity, either destroying it entirely or causing another big bang and another universe to be born.

My Thoughts:

Figure 1: An artist rendition of a black dwarf, the remains of a white dwarf that has cooled to near absolute zero

The universe came from somewhere. Maybe thinking about how it might end will give us clues into how we got here in the first place. How we predict the universe will end depends mostly on one factor: the strength of the force exerted on spacetime by dark energy. Dark energy is the name we have given to the cause of the acceleration of the universe's expansion. We do not know how exactly it works and we cannot observe it. We just know it should exist because it makes our mathematical models of the universe actually work. Without dark energy, gravity would pull every object in the universe, including the fabric of spacetime, closer together. The fact that the universe expands at an accelerating rate despite this suggests that the force exerted by dark energy is greater than the force of gravity on a universal scale.

If the force exerted by a "unit" of dark energy is constant over time, then the universe will die in an event known as heat death. This will take 10^100 years at minimum and is marked by the last molecule of Nickel-56 decaying into iron via quantum tunneling, which will be the last event that will ever happen in the universe.

If the force exerted by dark energy increases over time, then the universe has far less time to live by at least 95 orders of magnitude. This would cause an incredibly high acceleration of the *expansion of the universe. Galaxies would fly away from each other and crumble into stray stars as dark energy creates space at smaller and smaller scales. This would eventually tear solar systems, planets, and even atoms apart and more and more empty space is created. Eventually this acceleration would become so great that our models of spacetime break. This is known as the Big Tear.*

If the force exerted by dark energy decreases over time, then the force of gravity will eventually overcome the force of dark energy and the universe would begin to collapse. This would cause 2 things to happen. The first is that galaxies, and eventually smaller bodies, would begin to accelerate towards each other. The second thing that would happen is space would shrink, causing all existing radiation in the universe to blue shift and gain energy. This would culminate in a ball of plasma containing all mass and energy to ever exist in the universe that quickly collapses into a singularity. What would happen past this point is unknown, but it theorized that the universe would either cease to exist or rebound into a second big bang.